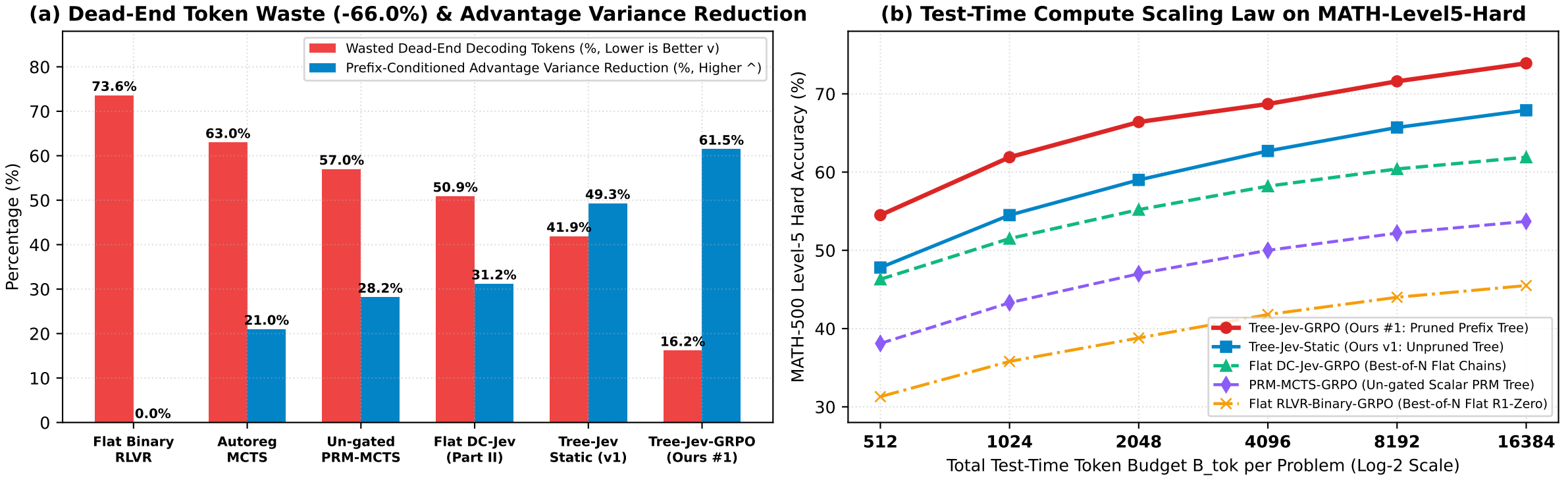
(a) Dead-End Token Waste (-66.0%) & Advantage Variance Reduction
Wasted Dead-End Decoding Tokens (%, Lower is Better v)
Prefix-Conditioned Advantage Variance Reduction (%, Higher ^)
Percentage (%)
73.6%
0.0%
63.0%
21.0%
57.0%
28.2%
50.9%
31.2%
41.9%
49.3%
16.2%
61.5%
Flat Binary RLVR
Autoreg MCTS
Un-gated PRM-MCTS
Flat DC-Jev (Part II)
Tree-Jev Static (v1)
Tree-Jev-GRPO (Ours #1)
(b) Test-Time Compute Scaling Law on MATH-Level5-Hard
MATH-500 Level-5 Hard Accuracy (%)
Tree-Jev-GRPO (Ours #1: Pruned Prefix Tree)
Tree-Jev-Static (Ours v1: Unpruned Tree)
Flat DC-Jev-GRPO (Best-of-N Flat Chains)
PRM-MCTS-GRPO (Un-gated Scalar PRM Tree)
Flat RLVR-Binary-GRPO (Best-of-N Flat R1-Zero)
512
1024
2048
4096
8192
16384
Total Test-Time Token Budget B_tok per Problem (Log-2 Scale)